

Serhan YILMAZ

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EDUCATION

Computer Science — Bachelor of Science
Sabanci University

Sep 2021 - Jun 2025
cGPA: 3.59/4

- Ranked top 0.02% among 2.9 million students in the National University Entrance Exam

PUBLICATIONS

- [1] **Serhan Yilmaz** and Kemal Oflazer, "CASCADES: Compound AI Systems for Controlled And Diverse Question Generation," in preparation, 2025.
- [2] Angelika Romanou, [...], **Serhan Yilmaz**, [...] Sara Hooker, Antoine Bosselut, "INCLUDE: Evaluating Multilingual Language Understanding with Regional Knowledge," **ICLR 2025 - Spotlight**.

WORK AND RESEARCH EXPERIENCE

Stealth Startup — Founding Engineer

Mar 2025 - Present

- Building AI-powered financial tools for enterprise — US-based stealth startup
- Leading **agentic AI** development, building **full-stack** web apps, and database **ETL** processes

Carnegie Mellon University — Research Internship

Dec 2023 - Present

- Research internship with Professor Kemal Oflazer at Carnegie Mellon University on advanced question generation and multi-agent frameworks
- Developed a novel **Compound AI system** utilizing multiple specialized language models in structured workflows for enhanced question generation
- Implemented sophisticated **LLM-as-a-Judge** methodologies with Chain-of-Thought reasoning for semantic and structural assessment
- Designing iterative feedback loops in multi-agent collaboration frameworks, achieving over **60% improvement** over baseline generation methods
- Authored the CASCADES framework paper, introducing compound AI systems for controlled question generation through structured LLM orchestration and iterative refinement

Expedition Aya - Cohere for AI — Research Contributor

Aug 2024 - Oct 2024

- Co-authored INCLUDE, a multilingual benchmarking dataset for LLMs (ICLR 2025 submission)
- Built **Python** pipelines using LLM and VLMs for automated extraction of multichoice questions from educational resources across diverse languages
- Involved in dataset curation with **1000+ questions** and mentoring international contributors to increase diversity

TU Darmstadt — Research Intern

Jul 2024 - Sep 2024

- Developed a self-sustaining agentic chatbot system for psychological assessment of Ukrainian refugees under Prof. Iryna Gurevych
- Implemented complex decision-making pipelines for autonomous question planning, contextual understanding, and verdict generation
- Optimized the system for deployment at Charité Berlin, integrating cultural sensitivity parameters and robust assessment protocols

KTH Royal Institute of Technology — Research Intern

Jun 2024 - Sep 2024

- Working on improving voice activity prediction models under Prof. Gabriel Skantze to create conversational systems capable of human-like turn-taking dialogue
- Implemented key-value caching for real-time inference and experimented with attention sinks for higher accuracy while maintaining efficiency
- Implemented data augmentation techniques for audio processing, enhancing model robustness in real-world scenarios
- Successfully deployed enhanced models on **Furhat Robotics'** humanoid robots, improving conversational turn-taking accuracy by **93%**

Yapi Kredi Technologies — NLP R&D Engineering Intern **Oct 2023 - Apr 2024**

- Enhanced document interpretation algorithms using OCR-free **Donut Transformers**
- Fine-tuned language models, reducing annotation effort by **40%** through transfer learning

EPFL — Summer@EPFL Intern **Jun 2023 - Sep 2023**

- Initiated Project Charisius, for developing robust, privacy-preserving Federated ML environments
- Optimized gradient aggregators, leveraging **PyTorch** and **CUDA** to achieve performance improvements of **2200%** compared to previous versions
- Published the Charisius library for accelerated Statistics and Federated ML applications, reducing runtime by up to **96%**
- Developed documentation website and Flask-backed benchmarking leaderboard for the project

Sabanci University — Undergraduate Teaching Assistant **Sep 2022 - Jan 2023**

- Wrote homework assignments for students and held weekly recitation and office hours

Boston University — Research Assistant **Jun 2022 - Sep 2022**

- Conducted research on Threat Modeling and Component Design for large-scale server systems under Prof. Rabia Tugce Yazicigil
- Developed a **C++** tool providing **336 combinations** to design secure servers for resource-intensive applications

Pharus Tech — Intern **Jul 2020 - Sep 2020**

- Analyzed and visualized data from the ESCO dataset, with information on **3008 occupations** and **13890 skills**
- Developed data processing pipelines, improving data analysis efficiency by **30%**

LEADERSHIP EXPERIENCE

kAi Sabanci — Founder & President **Mar 2023 - Present**

- Founded and led Sabanci University's AI club, growing membership to **600+** with **80% active participation**
- Organized **17 events** in one term, with workshops on Gen AI, NLP, ML, and DL, serving as a voluntary TA
- Secured kAi Sabanci's position as a founding member of the **NVIDIA Student Network**
- Facilitated high-profile visits, including hosting NVIDIA AI chief Simon See and organizing a visit to the Turkish Presidency's Digital Transformation Office
- Supervised **6 major projects**, including ChatSU, and NeRF campus modeling
- Selected for the prestigious **NVIDIA Student Spotlight** series for outstanding leadership and achievement

AWARDS

Sakip Sabanci Award for Outstanding Success, Sabanci University **Sep 2024**

- Demonstrated academic excellence throughout the year. Awarded **50% tuition scholarship**

Sakip Sabanci Award for Outstanding Success, Sabanci University **Sep 2022**

- Ranked top **4%** in class, demonstrating academic excellence. Awarded **100% tuition scholarship**

Innovation in Crisis Award, The Royal Swedish Academy of Engineering Sciences **May 2020**

- Secured **1st place** as part of team United in Crisis for innovative solutions during challenging times

Junior Academy Membership, The New York Academy of Sciences **Sep 2019**

- Selected for "The Junior Academy" program with a highly competitive **8% acceptance rate**

SKILLS

Languages & Tools: **Python, Docker, Linux, AWS, GCP, Git, Javascript, C, C++**

Libraries & Frameworks: **PyTorch, Transformers, LangChain, OpenCV, PostgreSQL, React**